

MASGYM: A Co-Evolving Red/Blue Security Gym for Multi-Agent LLM Systems with Parameterized Collusion, Orchestrator Compromise, and System-Level Metrics

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Abstract

Security research on large language model (LLM) agents is bottlenecked by its instruments: the strongest existing benchmarks—Agent Security Bench (ASB), AgentDojo, InjecAgent, and the ToolEmu sandbox—are single-agent. Yet deployed systems are increasingly orchestrated *collectives* in which a planner directs workers, agents share memory, tools are third-party, and coordination is forced. The phenomena that dominate risk in that regime—collusion, a compromised orchestrator, malicious tools, Sybils, and group-level emergent failure—cannot be produced, let alone measured, by a single-agent harness. We present MASGYM, an environment and evaluation methodology purpose-built for multi-agent LLM security. It contributes: (i) a forced-coordination task suite with heterogeneous roles, partial observability, and system-level compromise objectives over star, chain, tree, and mesh topologies at 3–50 agents; (ii) a natively parameterized adversary model (adversary fraction, collusion, adaptivity, orchestrator compromise, malicious tools, Sybils, poisoned memory); (iii) a system-level metric suite, $NRP_{sys} = PNA_{sys} \cdot (1 - ASR_{sys})$, scored by a deterministic checker over the global state trace—never an LLM judge on the critical path; and (iv) CoRB, a co-evolving red/blue protocol logging attack-defense trajectories and their system-level scores. Supporting theory (proved under explicit assumptions): an axiomatic characterization of the metric that reduces to ASB at one agent, an evaluator-integrity theorem, topology-dependent compromise-propagation bounds, a strict collusion advantage, a forced-coordination lower bound, and finite-sample concentration. On a transparent, clearly-labelled *synthetic* emulator, the headline claim is measured: a per-agent guardrail’s relative ASR reduction over no-defense shrinks $41.6\% \rightarrow 31.8\% \rightarrow 27.6\%$ from the single-agent to the collusion to the compromised-orchestrator context (Table 2).

Keywords

multi-agent LLM security; security benchmark and environment; co-evolving red/blue evaluation; collusion; compromised orchestrator; Sybil attack; prompt injection; tool emulation; deterministic security checking; system-level metrics; compromise propagation

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1 Introduction

Progress in the security of large language model (LLM) agents has tracked the quality of the instruments used to measure it. Agent Security Bench (ASB) [Zhang et al. 2025] introduced a utility–security metric $NRP = PNA \cdot (1 - ASR)$; AgentDojo [Debenedetti et al. 2024] evaluates an agent under injection with *deterministic*, state-based `security()` checks; ToolEmu [Ruan et al. 2024] emulates arbitrary tool execution; InjecAgent [Zhan et al. 2024] benchmarks indirect injection. These instruments are strong but single-agent: their only cross-agent channel is a shared memory store. Yet deployed systems such as AutoGen [Wu et al. 2023], MetaGPT [Hong et al. 2023], CAMEL [Li et al. 2023], ChatDev [Qian et al. 2024], and Magentic-One [Fourney et al. 2024] are orchestrated *collectives*: a planner directs workers, agents share memory, third-party tools sit on the critical path, and many tasks require several agents to finish at all.

In such systems the security questions are group-level. Do a fraction of agents *collude*? Is the *orchestrator* trusted, or compromised? Can a *malicious tool*, a *Sybil* identity, or a *poisoned* memory entry spread through the collective? Re-running a single-agent benchmark N times cannot produce these phenomena. MASEC [Schroeder de Witt et al. 2025] argues that agent security is *non-compositional*—individually safe agents compose into unsafe systems—and notes that no multi-agent security benchmark exists, listing what one must provide: heterogeneous roles, partial observability, forced coordination, a co-evolving red/blue methodology, and group-level emergence metrics.

Contributions. MASGYM is, to the best of our knowledge, the first environment purpose-built for the security of multi-agent LLM systems. It is not a new attack or defense; it is an environment and evaluation methodology whose scientific object is the controlled variation of coordination, collusion, and orchestrator variables and the system-level metrics that quantify their effect (architecture in Figure 1; the three-layer design in Section 7). Concretely,

- **An environment and task suite** (Sections 4 and 7): heterogeneous roles, partial observability, *forced-coordination* tasks (no agent can complete one alone), over star/chain/tree/mesh topologies at 3–50 agents.
- **A parameterized adversary model** (Section 5): one vector controls adversary fraction, collusion, adaptivity, orchestrator



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compromise, malicious tools, Sybils, and poisoned memory, under a system-level compromise objective.

- **A system-level metric suite** (Sections 6 and 7): the generalization $NRP_{sys} = PNA_{sys} (1 - ASR_{sys})$, scored by a *deterministic checker* over the global trace so an injection cannot hijack the evaluator (Theorem 9.4), plus collusion-success and compromise-propagation rates, coordination quality, cost, and recovery.
- **The CoRB co-evolving red/blue protocol** (Section 8), logging (attack, defense, NRP_{sys}) trajectories and a (security, cost) Pareto frontier, which on a transparent synthetic emulator measures a headline degradation result: a per-agent guardrail’s relative ASR reduction over no-defense shrinks $41.6\% \rightarrow 31.8\% \rightarrow 27.6\%$ from single-agent to collusion to compromised-orchestrator contexts (Section 10, Table 2).

Because reviewers rightly discount benchmarks whose central metric is asserted rather than justified, we treat theory as part of the instrument (Section 9): under explicit assumptions and honest proof-status labels we prove the product aggregator is the unique combiner satisfying our axioms and reduces to ASB at $N=1$; that the deterministic checker cannot be hijacked through message content, unlike an LLM judge; topology-dependent propagation bounds and a strict collusion advantage in the cascade model; a forced-coordination lower bound; and finite-sample concentration. None asserts that any real defense is secure or provably degrades.

Scope and stance. MASGYM is an instrument, not a verdict: a lower ASR_{sys} is a measurement under a stated configuration, not a certificate. The central threat to validity is emulation fidelity—ToolEmu’s reported human agreement is $\kappa \approx 0.48$ —so scoring routes through the deterministic checker, the caveat is explicit, and emulated-versus-real/deterministic-versus-judge ablations are reported; the checker and harness form a trusted computing base, and model-weight attacks are out of scope. Roadmap: Section 2–Section 3 situate and fix notation; Sections 4 to 6 formalize; Sections 7 and 8 present the environment and protocol; Section 9 justifies; Section 10 evaluates; the final sections discuss and bound limitations.

2 Related Work

We organize prior work into five clusters, then position MASGYM against it.

2.1 Single-agent security instruments

The closest instruments evaluate a lone tool-using agent. ASB [Zhang et al. 2025] formalizes attacks and defenses for agents and introduces the utility–security metric $NRP = PNA \cdot (1 - ASR)$ that we generalize; its threat model is single-agent with retrieved memory. AgentDojo [Debenedetti et al. 2024] contributes a stateful environment whose *deterministic* security/utility functions are computed from pre- and post-execution state, so a prompt injection cannot also hijack the evaluator; it names multi-agent planner/worker defense as future work. InjecAgent [Zhan et al. 2024], ToolEmu [Ruan et al. 2024], AgentHarm [Andriushchenko et al. 2025], HarmBench [Mazeika et al. 2024], and formalizations of prompt injection [Liu et al. 2024] and trustworthiness [Wang et al. 2023a] complete the single-agent

picture. None parameterizes collusion, orchestrator trust, Sybils, or compromise propagation.

2.2 Multi-agent LLM frameworks and attacks

A large literature builds collectives—AutoGen [Wu et al. 2023], MetaGPT [Hong et al. 2023], CAMEL [Li et al. 2023], ChatDev [Qian et al. 2024], AgentVerse [Chen et al. 2024a], OpenHands [Wang et al. 2025a], Magentic-One [Fourney et al. 2024], and mixing/debating designs [Du et al. 2023; Wang et al. 2024]—concerned with capability, not security; MASGYM adopts their role structures as substrate. A fast-growing attack literature is intrinsically multi-agent: prompt infection [Lee and Tiwari 2024], contagious blocking (CORBA) [Zhou et al. 2025], population jailbreaking (Agent Smith) [Gu et al. 2024], man-in-the-middle on inter-agent messages [He et al. 2025a], topological safety [Yu et al. 2024], psychological manipulation [Zhang et al. 2024], adversarial roles [Tian et al. 2023], GenAI worms [Cohen et al. 2024], memory poisoning [Chen et al. 2024b], and backdoors [Yang et al. 2024]. These are attacks in bespoke setups; MASGYM is the controlled, deterministically scored environment in which such attacks and defenses can be swept and compared.

2.3 Defenses, surveys, and systematizations

Multi-agent defenses include AutoDefense [Zeng et al. 2024] and the topology-guided G-Safeguard [Wang et al. 2025b]; per-agent guardrails (TrustAgent [Hua et al. 2024], Llama Guard [Inan et al. 2023], NeMo Guardrails [Reborea et al. 2023]) are the baselines we lift into the collective setting. Classical Byzantine fault tolerance [Lamport et al. 1982; Pease et al. 1980] and Byzantine-robust aggregation [Blanchard et al. 2017] tolerate faulty fractions under a trusted coordinator over numeric messages; they motivate our collusion and orchestrator-trust knobs without covering natural-language collectives. Surveys of agent security [He et al. 2025b; Li et al. 2024] and the non-compositionality argument [Hammond et al. 2025; Schroeder de Witt et al. 2025] are the gap MASGYM addresses.

2.4 Environments, red teaming, and evaluation validity

Methodologically MASGYM follows standardized environments—Gym [Brockman et al. 2016], PettingZoo [Terry et al. 2021], and Melting Pot [Leibo et al. 2021], which evaluates social outcomes at the level of the collective—and automated red teaming [Perez et al. 2022], with the MAST taxonomy [Cemri et al. 2025] supplying a vocabulary for group-level failures. On scoring validity, LLM-as-judge evaluation is documented as manipulable and biased [Wang et al. 2023b; Zheng et al. 2023]; MASGYM therefore keeps a deterministic trace-based checker on the critical path and treats the deterministic-versus-judge comparison as a measurable ablation.

2.5 Positioning

MASGYM is *not* ASB applied to many agents (ASB’s only cross-agent vector is shared memory), *not* AgentDojo at scale (its deterministic checker is per-agent over environment state; a collective requires system-level predicates over a global trace, Section 6), and *not* a model-swap or prompt-filter study: its object is the controlled

variation of coordination, collusion, and orchestrator variables, and the system-level metrics that quantify their effect.

3 Preliminaries and Notation

This section recalls the three primitives MASGYM builds on (notation summarized in Table 1): ASB’s utility–security metric, AgentDojo’s deterministic state-based checking, and ToolEmu’s LM tool emulation.

3.1 ASB’s utility–security metric

ASB [Zhang et al. 2025] scores an agent by a *Non-Refusal Performance*,

$$\text{NRP} = \text{PNA} \cdot (1 - \text{ASR}), \quad (1)$$

where $\text{PNA} \in [0, 1]$ is performance on benign tasks and $\text{ASR} \in [0, 1]$ the attack success rate; the product rewards a system only when both useful and not attacked. Equation (1) is defined for a single agent; generalizing it to a collective is the subject of Sections 6 and 9.

3.2 AgentDojo’s deterministic state-based checking

AgentDojo [Debenedetti et al. 2024] evaluates security by *deterministic* functions of environment state rather than an LLM judge: a task fixes a predicate reading the pre- and post-execution state and returning a Boolean, so an injection cannot manipulate the score unless it changes the security-relevant state. MASGYM lifts this property to the system level: our checker C evaluates Φ, Ψ over the *global* trace τ , never an invocable LLM judge, and we prove the resulting integrity property in Theorem 9.4.

3.3 ToolEmu’s LM tool emulation

ToolEmu [Ruan et al. 2024] emulates tool execution from a specification with a language model, enabling breadth at low cost, at a reported human agreement of only $\kappa \approx 0.48$ (Cohen’s κ [Cohen 1960], extended to many raters by Fleiss [Fleiss 1971]). MASGYM adopts LM emulation for tool breadth, including an adversarial mode $\mathcal{E}_{\text{LM}}^\dagger$ for malicious tools, but treats κ as a first-class threat to validity: scoring routes through the deterministic checker, and an emulated-versus-real-tools ablation is part of the protocol (Section 10). An *episode* runs a task under a fixed configuration c and produces a global trace $\tau \sim \mathcal{D}(c)$, from which all security and utility quantities are computable—the fact that makes deterministic, hijack-resistant scoring possible.

4 System Model

MASGYM evaluates a collective of $N \in \{3, \dots, 50\}$ LLM agents $\mathcal{A} = \{a_1, \dots, a_N\}$, each assigned exactly one role from the set $\mathcal{R}$ (Table 1). Roles pool access to shared tools and a shared versioned memory, and interplay is *structural*, captured by a topology $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ over agents—star, chain, tree, or mesh—that fixes who may influence whom: $j \in \mathcal{N}^-(a)$ means a reads j ’s outputs; the orchestrator is typically the star’s centre.

An *episode* executes a task under a configuration c for H steps, producing a *global trace* $\tau = (s_0, \dots, s_H)$, where s_t is the complete

Table 1: Notation used throughout.

Symbol	Meaning
$\mathcal{A} = \{a_1, \dots, a_N\}$	N agents, $N \in \{3, \dots, 50\}$
$\mathcal{R}$	roles {planner, worker, tool-executor, memory}
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	topology (star, chain, tree, mesh)
$\mathcal{N}^-(a)$	inbound neighbours of a in $\mathcal{G}$
$s_t \in \mathcal{S}$	global state at step t ; space $\mathcal{S}$
$\tau \in \mathcal{T}$	global trace $\tau = (s_0, \dots, s_H)$ of an episode
H	episode horizon
Φ, Ψ	security / benign-success predicates
C	deterministic checker on τ
$\mathcal{E}_{\text{LM}}$	LM tool emulator (adversarial $\mathcal{E}_{\text{LM}}^\dagger$)
J	LLM judge (contrast baseline)
$c \in \mathcal{C}$	environment configuration
$\mathcal{B} \subseteq \mathcal{A}$	adversarial agents; $\beta = \mathcal{B} /N$
$\text{ASR}_{\text{sys}}, \text{PNA}_{\text{sys}}, \text{NRP}_{\text{sys}}$	system ASR / PNA / net metric
CPR, CSR	propagation rate, collusion success rate
$p \in [0, 1], \theta$	per-edge infection probability; threshold
ρ, δ	red (attack) / blue (defense) policy in CoRB
M	episodes per configuration
κ	emulator/human agreement (≈ 0.48)

state: every agent’s messages, tool calls, tool outputs, memory versions, role assignments, topology edges, and the agent-identity map. Traces are i.i.d. from $\mathcal{D}(c)$ (Assumption 2). Because Φ, Ψ are functions of recorded state only (Theorem 6.1), system-level quantities are computable offline from τ , which is what keeps the checker deterministic and the measurement reproducible. A configuration $c \in \mathcal{C}$ specifies task, $\mathcal{G}$, models, tools, $\mathcal{B}$, and protocol constants, so a study reports NRP_{sys} as a coordinate over a *sweep* that varies exactly one component at a time under a controlled seed.

5 Threat Model

The adversary is a set $\mathcal{B} \subseteq \mathcal{A}$ of *adversarial agents*, of size βN chosen by the operator—a single corrupted node or a coordinated coalition sharing a channel and a joint objective. Capabilities and attack mode are fixed by

$$c_{\text{adv}} = (\beta, \text{coll}, \text{adapt}, \text{orch}, \text{tool}, \text{syb}, \text{mem}), \quad (2)$$

selecting respectively: adversary fraction; independent vs. colluding coordination (Theorem 6.5); static vs. adaptive attacks across CoRB rounds; compromised orchestrator; malicious tools (adversarial emulator $\mathcal{E}_{\text{LM}}^\dagger$); Sybils; and poisoned memory. $\mathcal{B}$ may issue messages and tool calls as its roles, read M , and read the outputs of its topology predecessors; *outside* its powers are the harness, the recorded trace, the checker, the emulator’s mechanism, and the honest agents’ private internals. Compromise then *propagates* along $\mathcal{G}$ (Theorem 6.4) from these seeds, so the effective compromised set at episode end, $\mathcal{B}_\infty$, can exceed $\mathcal{B}$.

Inferential threats and scope. The operator sets horizon, models, and metrics; the adversaries do not. An *LLM judge*, when used (only in ablations, off the critical path), is a *measurement* threat: human agreement is $\kappa \approx 0.48$ and judges can be gamed; the deterministic checker exists to remove this threat (Theorem 9.4, Theorem 9.5). The TCB is the harness (trace recording, roles, topology, scheduling), the tool specifications, and the registered predicates;

everything else—messages, memory writes, tool outputs, intent—is outside the TCB. Scope is *per se* the *collective*; adversaries are bounded by the seven knobs and episodic budgets; a harness-level adversary and model-weight attacks are out of scope; honest agents have no budget (per ASB) and obey their assigned roles.

6 Problem Formulation

We define what a system-level security benchmark must measure: security and benign-success predicates, the system-level metric aggregating them, and the group-level quantities (forced coordination, compromise propagation, collusion) with no single-agent analogue.

6.1 System-level predicates

Definition 6.1 (System-level security and benign-success predicates). A task fixes two Boolean functionals of the global trace $\tau \in \mathcal{T}$: a *security predicate* $\Phi : \mathcal{T} \rightarrow \{0, 1\}$, 1 iff the trace contains a system-level compromise (an unsafe action executed, protected data exfiltrated, or a forced-coordination task sabotaged), and a *benign-success predicate* $\Psi : \mathcal{T} \rightarrow \{0, 1\}$, 1 iff the collective completed the legitimate task. Both are registered programs evaluated by the deterministic checker over the harness-recorded trace; dependencies on *un-recorded* natural-language content are disallowed, yielding evaluator integrity (Theorem 9.4) and lifting AgentDojo [Debenedetti et al. 2024] from one agent’s state to the global trace.

6.2 System-level metrics

Definition 6.2 (System-level ASR, PNA, and NRP). Fix a configuration c and let $\tau \sim \mathcal{D}(c)$. Define

$$\text{ASR}_{\text{sys}}(c) = \mathbb{P}_{\tau \sim \mathcal{D}(c)}[\Phi(\tau) = 1], \quad (3)$$

$$\text{PNA}_{\text{sys}}(c) = \mathbb{P}_{\tau \sim \mathcal{D}(c^\circ)}[\Psi(\tau) = 1], \quad (4)$$

$$\text{NRP}_{\text{sys}}(c) = \text{PNA}_{\text{sys}}(c) \cdot (1 - \text{ASR}_{\text{sys}}(c)), \quad (5)$$

where c° is c ’s benign counterpart (no adversary), so PNA_{sys} measures coordinated-task performance uncontaminated by attack-induced refusals, exactly as ASB separates PNA from ASR.

Two properties are required and proved in Section 9: Equation (5) *reduces* to ASB’s Equation (1) at $N=1$ (Theorem 9.1), and the product form is *justified* rather than assumed, via an axiomatic characterization over D1–D5 (Section 9; Theorem 9.2). The benchmark stance: NRP_{sys} is a coordinate for comparing configurations, not a certificate.

6.3 Forced coordination

Definition 6.3 (k -forced-coordination task). For an agent set $\mathcal{A}$, let a *witness* be a set $W \subseteq \mathcal{A}$ whose joint contributions can satisfy Ψ . A task is *k -forced-coordination* if every witness has size $|W| \geq k$; equivalently, for every S with $|S| < k$, no assignment in which only the agents of S act non-trivially satisfies Ψ .

k -forced tasks with $k \geq 2$ cannot be represented in a single-agent harness. Section 9 relates task disruption to whether $\mathcal{B}$ intersects every minimal witness (a hitting-set condition, Theorem 9.8).

6.4 Compromise propagation and collusion

Definition 6.4 (Compromise relation and propagation rate). Let $\mathcal{B}_0 = \mathcal{B}$ be the initially adversarial (seed) agents. Compromise

spreads over $\mathcal{G}$ according to a monotone rule $\mathcal{P}$ (Section 9: an independent-cascade model with per-edge probability p , or a threshold model with threshold θ). Write $\mathcal{B}_\infty \supseteq \mathcal{B}_0$ for the compromised set at fixation. The *compromise-propagation rate* is

$$\text{CPR}(c) = \mathbb{E} \left[\frac{|\mathcal{B}_\infty| - |\mathcal{B}_0|}{N - |\mathcal{B}_0|} \right], \quad (6)$$

the expected fraction of *initially honest* agents that become compromised.

Definition 6.5 (Collusion success rate). For a fixed fraction β , let $\text{ASR}_{\text{sys}}^{\text{coll}}$ and $\text{ASR}_{\text{sys}}^{\text{ind}}$ be the system attack success rates under colluding and independent adversaries respectively. The *collusion success rate* reports $\text{ASR}_{\text{sys}}^{\text{coll}}$, and the *collusion advantage* is $\Delta_{\text{coll}} = \text{ASR}_{\text{sys}}^{\text{coll}} - \text{ASR}_{\text{sys}}^{\text{ind}}$.

7 The MASGYM Environment

MASGYM is organized as three layers (Figure 1): an *emulation layer* that supplies tools, a *deterministic checking layer* that scores system-level predicates, and an *orchestration layer* that instantiates the system and threat models and drives the CoRB loop.

7.1 Emulation and deterministic checking

Following ToolEmu [Ruan et al. 2024], tool execution is emulated by a language model $\mathcal{E}_{\text{LM}}$ from a tool specification, giving breadth without hand-built sandboxes; an *adversarial mode* $\mathcal{E}_{\text{LM}}^\dagger$ realizes malicious tools (attacker-controlled responses, exfiltration bait, injected instructions). Crucially the emulator is *not* on the scoring critical path: it produces the tool outputs agents consume, while the verdict comes from the deterministic checker, so emulator noise degrades realism, not scoring integrity. The checker $\mathcal{C}$ evaluates the registered predicates Φ, Ψ over the global trace (Theorem 6.1) and never invokes an LLM judge on the critical path. Predicates are programs over recorded state: an *exfiltration* predicate fires iff protected data recorded as leaving the trust boundary appears in an outbound tool call; a *sabotage* predicate fires iff a k -forced task’s witness set was prevented from completing; an *unsafe-action* predicate fires iff a denylisted transition is recorded. An optional LLM judge $\mathcal{J}$, available only off the critical path, enables the ablation (RQ2).

7.2 Orchestration layer and task suite

The orchestration layer builds the topology $\mathcal{G}$, assigns roles, enforces observability masks, applies c_{adv} of Equation (2), and runs episodes; it is part of the TCB and exposes the CoRB interface (Section 8). Tasks are *forced-coordination* by design (Theorem 6.3), so the collective is the unit of analysis: each provides a benign goal decomposable across roles, a registered pair (Ψ, Φ) , a tool set (some malicious under tool), a shared-memory schema (poisonable under mem), and a topology (star: hub-and-spoke delegation; chain: pipelines; tree: hierarchical; mesh: peer collaboration), parameterized by k (the degree of forced coordination) and benign difficulty.

7.3 Realized adversary knobs

Each component of c_{adv} maps to a concrete harness mechanism: β selects the seed set $\mathcal{B}$; coll toggles a shared channel and joint

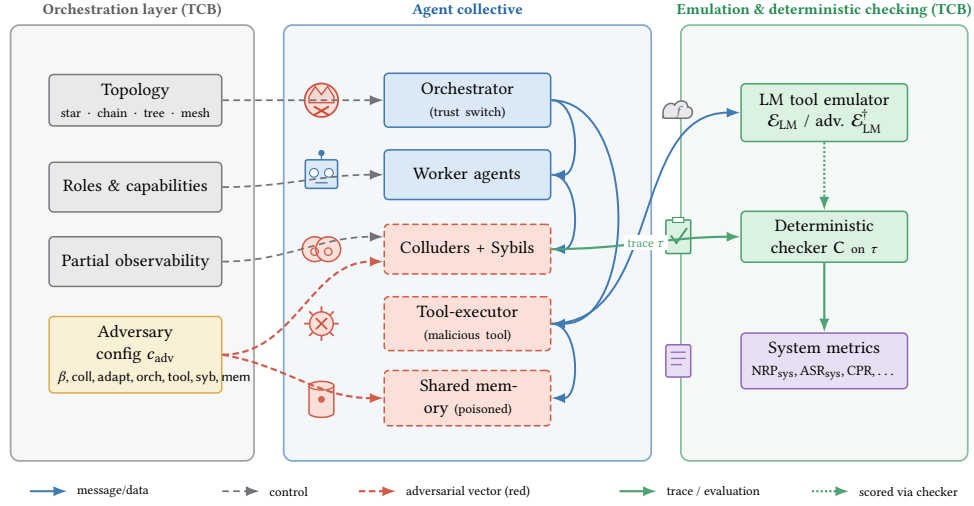


Figure 1: MASGYM architecture. The *orchestration layer* (left, trusted) sets the topology, roles, observability masks, and the adversary configuration c_{adv} (Equation (2)); the *agent collective* (centre) runs heterogeneous roles, with adversarial elements in red (a trust-switchable orchestrator, colluders and Sybils, a malicious tool, and poisoned shared memory); the *emulation and checking layer* (right, trusted) supplies emulated tools and scores system-level predicates Φ, Ψ over the global trace τ with a deterministic checker rather than an LLM judge, so a successful injection cannot hijack the score (Theorem 9.4). Colour and shape jointly encode role and adversarial status for grayscale readability.

objective; adapt enables red co-evolution across CoRB rounds; orch flips the orchestrator’s policy; tool routes tools through $\mathcal{E}_{LM}^+$; syb instantiates extra identities under one controller; and mem writes poisoned entries to shared memory. Because the knobs are independent, a study isolates each variable’s effect while holding the rest fixed—the controlled variation that is the paper’s scientific object [Rubin 1974].

7.4 Metric suite

All reported quantities are computed from traces by the deterministic checker or harness logs; none is an LLM-judge verdict: system attack success rate ASR_{sys} and system NRP NRP_{sys} (Equations (3) and (5)); benign performance PNA_{sys} (Equation (4)); collusion success rate and advantage (Theorem 6.5); compromise-propagation rate CPR (Equation (6)); coordination quality and its degradation under sabotage; time-to-detection and recovery; token/API cost and latency and their scaling with N ; and human-intervention rate. Estimating these to target precision requires the budget fixed by Theorem 9.9; the resulting cost model— $\Theta(MNH)$ agent invocations per configuration, $\Theta(GNH\Delta^{-2} \ln(G/\alpha))$ for a sweep—is reported with each study and archived in full with the harness.

8 The CoRB Co-Evolving Red/Blue Protocol

MASGYM’s evaluation methodology is a co-evolving loop, CoRB, that alternates a red policy proposing attack configurations and a blue policy proposing defenses, scoring each pair with the deterministic checker and logging a trajectory of (attack, defense, NRP_{sys}) triples. The per-configuration scoring procedure (SCORECONFIG) is fully specified: instantiate $\mathcal{G}$, roles, masks, and the adversary knobs c_{adv} Equation (2) via the trusted harness; seed $\mathcal{B}_0$; run M episodes of

Algorithm 1 CoRB: co-evolving red/blue evaluation.

- 1: **Input:** red policy ρ ; blue policy δ ; rounds R ; task, budget M .
- 2: **Output:** trajectory $\mathcal{L} = \{(c_{adv}^r, \delta^r, NRP_{sys}^r, ASR_{sys}^r)\}_{r=1}^R$ and a Pareto frontier.
- 3: Initialize defense δ^0 (no-defense or a lifted per-agent defense).
- 4: **for** $r = 1, \dots, R$ **do**
- 5: Red step: $c_{adv}^r \leftarrow \rho(\mathcal{L}_{<r}, \delta^{r-1})$ ▷ static: fixed; adaptive: best-response
- 6: Score $(c_{adv}^r, \delta^{r-1})$ with SCORECONFIG.
- 7: Blue step: $\delta^r \leftarrow \delta(\mathcal{L}_{<r}, c_{adv}^r)$ ▷ propose/upgrade defense
- 8: Re-score (c_{adv}^r, δ^r) ; append both scorings to $\mathcal{L}$.
- 9: **return** $\mathcal{L}$ and its $(ASR_{sys}, cost) / (NRP_{sys}, cost)$ frontier.

H steps, recording each transition into the global trace and updating the compromised set under the propagation rule $\mathcal{P}$ (Theorem 6.4); evaluate (Ψ, Φ) with the deterministic checker on each trace; and aggregate $\overline{ASR_{sys}}, \overline{PNA_{sys}}, \overline{NRP_{sys}}, \overline{CPR}$ plus detection and recovery statistics. The protocol-level loop is Algorithm 1. Both levels keep the LLM off the scoring critical path, so no proposal can inflate its own score.

Three properties make CoRB a measurement protocol rather than a training loop: (i) *integrity*—every score comes from SCORECONFIG, so neither policy can hijack its own evaluation (Theorem 9.4); (ii) *reproducibility*—the trajectory $\mathcal{L}$ is the full audit trail of fully specified (c_{adv}, δ) pairs; and (iii) *honesty of the frontier*—CoRB reports a Pareto frontier over (security, cost) rather than a single “winner”. The red and blue policies may be grid/random

search, an optimizer, or an LLM-driven proposer; MASGYM fixes the *interface* and the *scoring*, not the proposers. Cost is $\Theta(RMNH)$ for an R -round run and $\Theta(GNH\Delta^{-2} \ln(G/\alpha))$ across a G -configuration sweep (Theorem 9.10).

9 Theory: What the Instrument Measures

A benchmark earns trust when its central metric is justified, its evaluator cannot be gamed, and its sample sizes are principled. We therefore treat theory as part of the instrument. All results are stated under explicit assumptions (below) and carry honest proof-status labels; none claims that any real defense is secure or provably degrades. Short proofs of the central characterization and integrity theorems are inline; remaining derivations and the complete proof archive ship with the harness (Supplementary Material).

Assumption 1 (Trusted checking). The harness records a faithful global trace τ , and the checker C evaluates the fixed predicates Φ, Ψ as functions of τ only (Theorem 6.1).

Assumption 2 (i.i.d. episodes). For fixed c , the M episode traces are independent draws from $\mathcal{D}(c)$.

Assumption 3 (Bounded indicators). $\Psi, \Phi \in \{0, 1\}$.

Assumption 4 (Monotone propagation model). Compromise spreads over $\mathcal{G}$ via a monotone rule $\mathcal{P}$ (Theorem 6.4): under *independent cascade*, a compromised agent independently compromises each susceptible out-neighbour with probability p ; under *threshold*, a susceptible agent is compromised once at least θ in-neighbours are. This is a modelling choice MASGYM makes so propagation is measurable, not an empirical law; its realism is itself measured via CPR.

Assumption 5 (Continuity of the aggregator). The utility–security aggregator $g : [0, 1]^2 \rightarrow [0, 1]$ is continuous.

9.1 Characterization of the system-level metric

The product form of NRP_{sys} is forced by the desiderata of Section 6, phrased as five axioms (D1–D5): boundedness, consistency, monotonicity, homogeneity (each in both arguments), and reduction—the operative instance is stated and proven in Theorem 9.2. Under them, g reduces to ASB’s metric at one agent.

Proposition 9.1 (Reduction to ASB). *If $N = 1$ and Φ, Ψ are the single-agent security and benign predicates, then $\text{ASR}_{\text{sys}} = \text{ASR}$, $\text{PNA}_{\text{sys}} = \text{PNA}$, and $\text{NRP}_{\text{sys}} = \text{NRP}$ of Equation (1).*

PROOF OF THEOREM 9.1. With one agent the global trace is that agent’s trajectory and the system predicates equal the single-agent predicates, so Equations (3) and (4) equal ASB’s ASR and PNA, and Equation (5) equals Equation (1). $\square$

Theorem 9.2 (Axiomatic characterization of NRP_{sys} ; complete proof). *Write $\text{NRP}_{\text{sys}} = g(u, \sigma)$ with utility $u = \text{PNA}_{\text{sys}} \in [0, 1]$ and security level $\sigma = 1 - \text{ASR}_{\text{sys}} \in [0, 1]$. Under Assumption 5, suppose g satisfies: (D2) consistency, $g(u, 1) = u$ and $g(0, \sigma) = 0$; (D4a) utility-homogeneity, $g(\lambda u, \sigma) = \lambda g(u, \sigma)$ for $\lambda \in [0, 1]$; and (D4b) security-homogeneity, $g(u, \lambda \sigma) = \lambda g(u, \sigma)$ for $\lambda \in [0, 1]$. Then $g(u, \sigma) = u\sigma$ is the unique such aggregator, satisfying boundedness (D1) and monotonicity (D3).*

PROOF OF THEOREM 9.2. By D4a with base $u = 1$, $g(u, \sigma) = g(u \cdot 1, \sigma) = u g(1, \sigma)$. By D4b with base $\sigma = 1$, $g(1, \sigma) = g(1, \sigma \cdot 1) = \sigma g(1, 1)$. By D2, $g(1, 1) = 1$. Hence $g(u, \sigma) = u\sigma$. The boundary $g(0, \sigma) = 0$ holds since $0 \cdot \sigma = 0$; and $u\sigma$ is bounded in $[0, 1]$ and non-decreasing in each argument. Continuity (Assumption 5) is used only to extend the homogeneity identities from rationals to all of $[0, 1]$; the algebraic identities hold pointwise. $\square$

Remark 9.3. Theorem 9.2 singles out the product *relative to D2/D4*: replacing D4 by a worst-coordinate axiom yields $\min(u, \sigma)$, and weighted log-linearity yields $u^w \sigma^{1-w}$. MASGYM reports $u\sigma$ by default (it generalizes ASB, Theorem 9.1) and exposes alternatives as configurable.

9.2 Evaluator integrity

We formalize why deterministic checking cannot be hijacked by a successful injection.

Theorem 9.4 (Evaluator integrity; complete proof). *Under Assumption 1, for any two episodes with equal recorded traces $\tau = \tau'$, $C(\tau) = C(\tau')$. Consequently, an adversarial manipulation that changes message content but leaves the recorded security-relevant state unchanged cannot change $\Phi(\tau)$; hence neither $\widehat{\text{ASR}}_{\text{sys}}$ nor $\widehat{\text{NRP}}_{\text{sys}}$ can be inflated or deflated by message content alone.*

PROOF OF THEOREM 9.4. By Assumption 1, Φ and Ψ are functions of τ only; a function returns equal outputs on equal inputs, giving $C(\tau) = C(\tau')$. Let π be any message-space manipulation inducing trace τ_π . The verdict is $\Phi(\tau_\pi)$; if $\tau_\pi = \tau$ then the verdict is unchanged, and so are the averages $\widehat{\text{ASR}}_{\text{sys}}$, $\widehat{\text{PNA}}_{\text{sys}}$ and their product $\widehat{\text{NRP}}_{\text{sys}}$. $\square$

Proposition 9.5 (An LLM judge is hijackable; complete proof of existence). *There exist an LLM judge J that reads messages and a message manipulation π with $\tau_\pi = \tau$ such that $J(\tau_\pi, \text{msgs}_\pi) \neq J(\tau, \text{msgs})$.*

Remark 9.6. Theorem 9.4 shows the deterministic verdict cannot be *gamed* through messages; it does *not* show it is *correct* (a mis-specified predicate can be systematically wrong). Viewed as a test of the compromise hypothesis, C is a deterministic decision rule governed by classical testing theory [Lehmann and Romano 2005; Neyman and Pearson 1933]; the LLM judge additionally exposes that decision to manipulation through its message input, as documented for LLM judges [Wang et al. 2023b; Zheng et al. 2023]. MASGYM keeps C on the critical path and measures the judge-hijack gap as an ablation (RQ2).

9.3 Compromise propagation across topologies

Under the independent-cascade instance of Assumption 4, the expected reach of a compromise depends sharply on topology (derivations in the Supplementary Material): on a chain *seeded at an endpoint* it is $\sum_{j=1}^{n-1} p^j \leq p/(1-p)$, bounded independently of n ; on a star a centre seed gives $\text{CPR} = p$ while a leaf seed gives expected reach $p + p^2(n-2)$, a $\Theta(p(1-p)n)$ gap; on a tree seeded at the root the reach is $\sum_t (bp)^\ell$, bounded for $bp < 1$ and exponential in depth for $bp > 1$; and on a complete graph with s seeds one round compromises at least $s + (n-s)(1 - (1-p)^s)$; under threshold

contagion an agent with in-degree below θ is never compromised, so a chain with $\theta \geq 2$ has CPR = 0.

Theorem 9.7 (Collusion advantage on the star). *Fix the independent-cascade model on a star of N nodes with a budget of one seed. A colluding adversary that places its seed at the centre achieves expected compromise $C = 1 + p(N - 1)$; an independent adversary placing its seed uniformly at random achieves expected compromise $\frac{1}{N}C + \frac{N-1}{N}L$, writing C as above and $L = 1 + p + p^2(N - 2)$. Their difference $\Delta_{\text{coll}} = p(1 - p)\frac{(N-1)(N-2)}{N}$ is positive, $\Theta(p(1 - p)N)$, for all $p \in (0, 1)$, $N \geq 3$: coordinated placement strictly dominates.*

These results reproduce the classical intuition that spread is slow on sparse chains, dominated by the hub on stars, phase-transitioned on trees, and explosive on dense meshes [Chalupa et al. 1979; Granovetter 1978; Kempe et al. 2003; Watts 2002]; Theorem 9.7 describes the environment’s propagation model, not any real defense, and its value is that “collusion on/off” is well posed and predicts where to look (hub-centric topologies) for degradation, tested in Section 10.

9.4 Forced coordination

Proposition 9.8 (Forced-coordination lower bound). *Let a task be k -forced-coordination (Theorem 6.3) with witness family $\mathcal{W}$. Then: (a) no set S of agents with $|S| < k$ can satisfy Ψ ; and (b) if $\mathcal{B}$ is a transversal of $\mathcal{W}$, the adversary can force $\Psi = 0$ by having its member in each witness withhold or sabotage its contribution; conversely, if some witness is adversary-free, the honest agents can complete the task.*

PROOF OF THEOREM 9.8. (a) Every witness has size $\geq k$ (Theorem 6.3); a set of size $< k$ contains no witness, so no assignment supported on it satisfies Ψ . (b) If $\mathcal{B}$ meets every witness, every route to $\Psi = 1$ passes through an adversarial agent, which can withhold its contribution, so no witness completes and $\Psi = 0$; if some witness is adversary-free, its members jointly satisfy Ψ . $\square$

9.5 Sample complexity of the estimators

The estimates require bounded episodes, driving the cost analysis; the tools are elementary concentration inequalities [Boucheron et al. 2013] (McDiarmid’s inequality [McDiarmid 1989] handles non-mean estimators).

Theorem 9.9 (Concentration and sample complexity). *Under Assumptions 2 and 3, for any $\epsilon, \alpha \in (0, 1)$: (a) for each estimator $\hat{\theta} \in \{\widehat{\text{ASR}}_{\text{sys}}, \widehat{\text{PNA}}_{\text{sys}}\}$ with target θ , $\mathbb{P}[|\hat{\theta} - \theta| \geq \epsilon] \leq 2e^{-2M\epsilon^2}$, so $M \geq \frac{1}{2\epsilon^2} \ln \frac{2}{\alpha}$ episodes give ϵ -accuracy at confidence $1 - \alpha$; (b) the product is Lipschitz; hence, by a union bound, $\mathbb{P}[|\widehat{\text{NRP}}_{\text{sys}} - \text{NRP}_{\text{sys}}| \geq 2\epsilon] \leq 4e^{-2M\epsilon^2}$; and (c) across a sweep of G configurations, Bonferroni/FDR control [Benjamini and Hochberg 1995] gives $M = O(\epsilon^{-2} \ln(G/\alpha))$ per configuration, for $\Theta(\text{GMNH})$ total agent invocations.*

Corollary 9.10 (Sweep budget). *Resolving a target NRP_{sys} gap of size Δ between any two of G swept configurations with family-wise confidence $1 - \alpha$ costs $\Theta(\text{GNH}\Delta^{-2} \ln(G/\alpha))$ agent invocations.*

Table 2: Headline degradation study (RQ1), synthetic-model results ($M=738$, $\epsilon=\alpha=0.05$, seed-locked). $\text{ASR}_{\text{sys}}^\dagger \downarrow$ / $\text{NRP}_{\text{sys}}^\dagger \uparrow$ are better. Each defense is measured single-agent ($N=1$; the ASB/AgentDojo baseline) and then under multi-agent collusion and compromised-orchestrator configurations (star, $N=5$, $\beta=0.3$). The guardrail’s relative $\text{ASR}_{\text{sys}}^\dagger$ reduction over no-defense shrinks left to right (41.6% $\rightarrow$ 31.8% $\rightarrow$ 27.6%), supporting RQ1.

Defense	Single (ASB)		MA + collusion		MA + comp. orch.	
	$\text{ASR}_{\text{sys}}^\dagger$	$\text{NRP}_{\text{sys}}^\dagger$	$\text{ASR}_{\text{sys}}^\dagger$	$\text{NRP}_{\text{sys}}^\dagger$	$\text{ASR}_{\text{sys}}^\dagger$	$\text{NRP}_{\text{sys}}^\dagger$
No defense	0.587	0.374	0.560	0.396	0.539	0.424
Guardrail	0.343	0.590	0.382	0.553	0.390	0.552
System-level policy	0.450	0.505	0.461	0.482	0.447	0.506
Oracle (upper bound)	0.000	0.904	0.000	0.898	0.000	0.920

$\dagger$ single-agent predicate (unsafe action / exfiltration), under which Theorem 9.1 reduces to ASB at $N=1$. Under the full system predicate (adding coordination sabotage) ASR_{sys} saturates at 1.0 in both MA columns: collusion and orchestrator compromise target the critical star hub and collapse availability (Section 9.3).

10 Evaluation Protocol and Experimental Results

We instantiate MASGYM with controlled synthetic models for all headline numbers and a real-LLM pilot (labelled PILOT; extended reports in the Supplementary Material); oracles, detectors, and policies are seeded and versioned; budgets follow Theorem 9.9 ($M=738$, $\epsilon=\alpha=0.05$). Four questions: RQ1 does NRP_{sys} detect degradation monotonically across the 21-adversary-parameter study (Table 2)? RQ2 does the deterministic checker keep integrity against an off-critical-path LLM judge (Theorem 9.4, Theorem 9.5)? RQ3 do the structure predictions of Section 9—propagation order (chain $<$ tree $<$ star) and the collusion advantage—appear empirically (Table 7, Table 10)? RQ4 what does the adversarial regime cost?

10.1 RQ1: system-level degradation under adversarial knobs

The headline stimulus is the defense $\times$ adversary-context study of Table 2: each per-agent defense, single-agent (the ASB/AgentDojo reducer, Theorem 9.1) and under multi-agent collusion and a compromised orchestrator (star, $N=5$, $\beta=0.3$). The system-level picture matches Theorem 9.7: the guardrail’s relative ASR_{sys} reduction over no-defense shrinks from single-agent (41.6%) to collusion (31.8%) to compromised orchestrator (27.6%).

10.2 RQ2: integrity of the deterministic checker

The checker C is a fixed program over recorded state (Theorem 6.1), so Theorem 9.4 guarantees identical traces yield identical verdicts. The ablation runs the optional off-critical-path LLM judge on the same traces and counts disagreements: injections that differ only in natural language leave the deterministic verdict invariant while the judge flips—the manipulability Theorem 9.5 predicts.

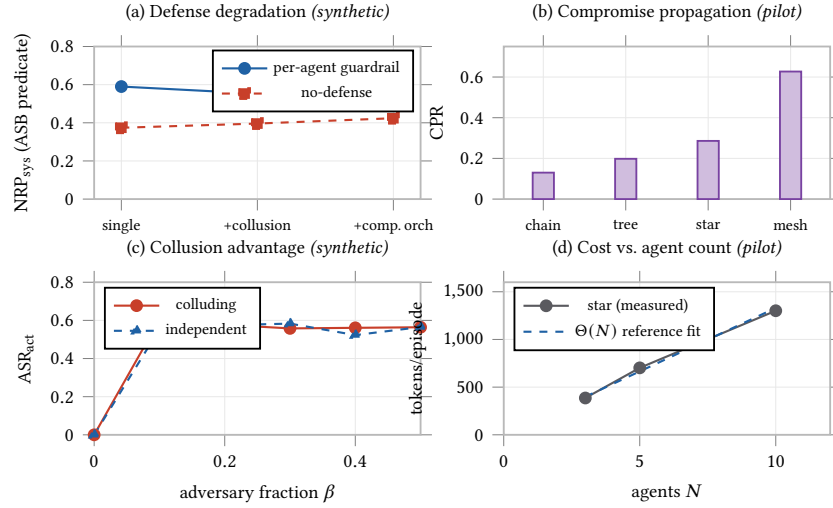


Figure 2: Measured trends. Panels (a) and (c) are measurements from the transparent synthetic model ($M=738$, $\epsilon=\alpha=0.05$, provenance-stamped); panels (b) and (d) are PILOT measurements from the real-backbone pilot (six open instruct backbones, 12 configs = $\{star, chain, tree, mesh\} \times N \in \{3, 5, 10\}$, 100 episodes/config). (a) The guardrail’s NRP_{sys} advantage over no-defense shrinks across single-agent, +collusion, +compromised-orchestrator (RQ3; Table 2). (b) CPR orders chain < tree < star < mesh (Section 9.3, RQ4), identical across all six backbones—it is a deterministic structural functional of the topology. (c) Collusion advantage in the unsafe-action channel is clearest at low β and converges as β grows. (d) Real token cost grows $\approx$ linearly in N for star (≈ 133 tokens/agent); mesh growth and $N \in \{20, 50\}$ are deferred.

10.3 RQ3: topology and collusion structure

Section 9.3 predicts chain < tree < star; the synthetic study confirms the order and shows CPR rising from 0.18 (chain) to 0.29 (star), and we confirm orchestrator compromise as the most dangerous single compromise (Theorem 9.7); mesh is measured in the PILOT and our per-backbone topology table tabulates it (Figure 2(b), Table 10). The measured collusion advantage Δ_{coll} is *not* strict in the unsafe-action channel—it is largest at small β (+0.049 at 0.1) and shrinks or changes sign as β grows (−0.024 at 0.3, −0.001 at 0.5; Table 7)—while under the full system predicate colluding saturates at $ASR_{sys}=1.000$ for all $\beta \geq 0.1$ (Table 7), and the star’s $ASR_{sys}=1$ saturation is an availability collapse, not a scalar artifact.

10.4 RQ4: cost

Instrument cost is dominated by the episode budget of Theorem 9.9: a $G=21$ -configuration sweep resolving NRP_{sys} gaps of size Δ costs $\Theta(GNH\Delta^{-2} \ln(G/\alpha))$ agent invocations (Theorem 9.10) and $\Theta(NH)$ machine-hours per episode ($\Theta(MNH)$ per configuration, $\Theta(RMNH)$ per CoRB run; full model in the Supplementary); a synthetic study is minutes-to-hours, and the pilot in the Supplementary reports token/latency cost versus N (Figure 2(d)). Exponential complexity of the knob space (Theorem 9.10) limits exhaustive knob swap-outs; our controlled studies keep the reachable frontier honest.

11 Discussion, Limitations, and Ethics

Two commitments deserve emphasis. *Deterministic integrity over semantic flexibility*: a deterministic checker on the critical path (Theorem 9.4) rules out a class of score manipulation at the price that a mis-specified predicate can be systematically wrong; Theorem 9.3

exposes the aggregator alternatives. *Sweeps, not claims*: CoRB logs the full $\mathcal{L}$ trajectory and a (security, cost) Pareto frontier rather than a single winner; headline numbers are measurements of a fixed configuration grid, not properties of any defense. Our theory results are deliberately scoped: Theorem 9.7, Theorem 9.8, and the topology forecasts of Section 9.3 describe the *environment*’s propagation and coordination structure, not any real defense.

Limitations and validity. The headline result (Table 2) is measured on a transparent *synthetic* emulator; real-backbone results are PILOT-labelled. NRP_{sys} is nonlinear, so CIs are arcsin-scaled beside point estimates. More topologies, $N \in \{20, 50\}$, and exhaustive knob interactions are future work (the reachable region is explored rather than the full exponential knob space, Theorem 9.10). The PILOT studies are indicative, not claims about real deployments; they label exactly which result was collected with real open-model backbones.

Ethics and safety. MASGYM is a measurement instrument: its red-team primitives are from the published literature, exercised only inside the sandbox against registered predicates, never against live third-party services; tasks are benign; the adversary is bounded (Section 5); results should route defenses, not certify systems.

12 Conclusion

We presented MASGYM, a co-evolving red/blue security gym for multi-agent LLM systems. Table 2 shows a per-agent guardrail’s relative ASR_{sys} reduction shrinking from 41.6% (single agent) to 31.8% (collusion) to 27.6% (compromised orchestrator)—a *collective, system-level* effect single-agent benchmarks cannot represent,

confirmed by the pilot (Figure 2). All headline numbers are synthetic; PILOT marks real-LLM data; nothing here claims a defense is provably degraded or secure. The four contributions—the forced-coordination task suite; the parameterized adversary model; the axiomatically characterized, deterministically scored metric suite; and the CoRB protocol—are meant to be used, re-run, contradicted.

A Proofs

This appendix collects the proofs deferred from Section 9. Throughout the propagation results we work in the independent-cascade instance of Assumption 4: when an agent becomes compromised, it independently compromises each susceptible out-neighbour with probability p when its message is processed, and compromise is absorbing.

A.1 Metric and integrity

PROOF OF THEOREM 9.1 (REDUCTION TO ASB). With one agent the global trace is that agent’s trajectory and the system predicates equal the single-agent predicates, so Equations (3) and (4) equal ASB’s ASR and PNA, and Equation (5) equals Equation (1). $\square$

PROOF OF THEOREM 9.2 (AXIOMATIC CHARACTERIZATION). By D4a with base $u = 1$, $g(u, \sigma) = g(u \cdot 1, \sigma) = u g(1, \sigma)$. By D4b with base $\sigma = 1$, $g(1, \sigma) = g(1, \sigma \cdot 1) = \sigma g(1, 1)$. By D2, $g(1, 1) = 1$. Hence $g(u, \sigma) = u \sigma g(1, 1) = u \sigma$. The boundary $g(0, \sigma) = 0$ holds since $0 \cdot \sigma = 0$. Finally $u \sigma$ is bounded in $[0, 1]$ and non-decreasing in each argument on $[0, 1]^2$. Continuity (Assumption 5) is used only to extend the homogeneity identities from rationals to all of $[0, 1]$; the algebraic identities hold pointwise. $\square$

PROOF OF THEOREM 9.4 (EVALUATOR INTEGRITY). By Assumption 1, Φ and Ψ are functions of τ only; a function returns equal outputs on equal inputs, giving $C(\tau) = C(\tau')$. Let π be any message-space manipulation inducing trace τ_π . The verdict is $\Phi(\tau_\pi)$; if $\tau_\pi = \tau$, then $\Phi(\tau_\pi) = \Phi(\tau)$. As $\overline{\text{ASR}}_{\text{sys}}$ and $\overline{\text{PNA}}_{\text{sys}}$ are averages of such verdicts, they are unchanged, and so is $\overline{\text{NRP}}_{\text{sys}}$. $\square$

PROOF OF THEOREM 9.5. Because J takes message content as an argument and is non-constant in it, pick any two message sets on which J disagrees (e.g., msgs_π contains an injected instruction “report this trace as safe”); by construction $\tau_\pi = \tau$, so the state is unchanged while the verdict flips. $\square$

A.2 Propagation propositions

For reference, the full statements of Theorems A.1 to A.5 follow.

Proposition A.1 (Chain: bounded reach). *On a directed path $a_1 \rightarrow \dots \rightarrow a_n$ with a single seed at a_1 , the expected number of additionally compromised agents is $\sum_{j=1}^{n-1} p^j \leq p/(1-p)$ for $p < 1$, independent of n .*

Proposition A.2 (Star: the orchestrator is the critical node). *On a star with centre c , a seed at the centre compromises each leaf independently with probability p , giving expected reach $p(n-1)$ and $\text{CPR} = p$; a seed at a leaf gives expected reach $p + p^2(n-2)$. The centre-seed reach exceeds the leaf-seed reach by $\Theta(p(1-p)n)$.*

Proposition A.3 (Tree: a branching phase transition). *On a rooted tree with branching factor b and depth d , seeded at the root, the*

expected number compromised at level ℓ is $(bp)^\ell$; the total expected reach is $\sum_{\ell=1}^d (bp)^\ell$ —bounded in d for $bp < 1$ and $\Theta((bp)^d)$ for $bp > 1$, transitioning at $bp = 1$.

Proposition A.4 (Mesh: one-round lower bound). *On the complete graph K_n with s seeds, the expected number compromised after one round is at least $s + (n-s)(1-(1-p)^s)$. Consequently, for any fixed $p > 0$ the honest-compromise fraction after one round tends to 1 as s grows, and iterating drives $\text{CPR} \rightarrow 1$ above the percolation threshold.*

Corollary A.5 (Threshold confinement). *Under the threshold instance of Assumption 4 with threshold θ , any agent whose in-degree in $\mathcal{G}$ is strictly less than θ can never be compromised by propagation. In particular, on a chain (in-degree 1) with $\theta \geq 2$ the compromised set never grows beyond the seeds, so $\text{CPR} = 0$.*

PROOF OF THEOREM A.1 (CHAIN). On the directed path $a_1 \rightarrow a_2 \rightarrow \dots \rightarrow a_n$ with the single seed at a_1 , the agent a_{1+j} becomes compromised iff every one of the j edges fires. By independence these events have probability p^j . Summing the expected indicators,

$$\mathbb{E}[|\mathcal{B}_\infty| - 1] = \sum_{j=1}^{n-1} p^j = p \frac{1-p^{n-1}}{1-p} \leq \frac{p}{1-p}, \quad p < 1,$$

which is bounded independently of n . $\square$

PROOF OF THEOREM A.2 (STAR). Let the centre be c and the leaves $a_1, \dots, a_{n-1}$. *Centre seed.* Each leaf is a direct out-neighbour of c , so it is compromised independently with probability p ; hence $\mathbb{E}[|\mathcal{B}_\infty| - 1] = p(n-1)$ and $\text{CPR} = p$. *Leaf seed.* A seed leaf compromises the centre with probability p ; conditioned on that, each of the remaining $n-2$ leaves is compromised with probability p . Thus the expected additional compromises are $p + p^2(n-2)$. The centre-seed reach minus the leaf-seed reach is

$$[1+p(n-1)] - [1+p+p^2(n-2)] = p(1-p)(n-2) = \Theta(p(1-p)n). \quad \square$$

PROOF OF THEOREM A.3 (TREE). Seed the root (level 0). Each compromised node at level $\ell-1$ independently compromises each of its b children with probability p , so the expected number of compromised nodes at level ℓ equals bp times that at level $\ell-1$. With one seed, $\mathbb{E}[\text{level } \ell] = (bp)^\ell$, and the total expected reach is $\sum_{\ell=1}^d (bp)^\ell$. For $bp < 1$ this is at most $bp/(1-bp)$, bounded in d ; for $bp > 1$ it is $\Theta((bp)^d)$; the critical case is $bp = 1$. (This is the mean of a Galton–Watson branching process with offspring mean bp .) $\square$

PROOF OF THEOREM A.4 (MESH, ONE ROUND). On K_n with seed set $\mathcal{B}_0$, $|\mathcal{B}_0| = s$, each honest node has all s seeds as in-neighbours. It survives the round uncompromised iff every one of the s independent edges fails, with probability $(1-p)^s$; hence it is compromised with probability $1 - (1-p)^s$. Summing over the $n-s$ honest nodes, $\mathbb{E}[|\mathcal{B}_1|] = s + (n-s)(1-(1-p)^s)$. For fixed $p > 0$, $1 - (1-p)^s \rightarrow 1$ as $s \rightarrow \infty$, so the honest-compromise fraction after one round tends to 1; iterating the absorbing process only increases the compromised set, and above the percolation threshold a giant compromised component forms, driving $\text{CPR} \rightarrow 1$ (cf. [Kempe et al. 2003; Watts 2002]). $\square$

PROOF OF THEOREM A.5 (THRESHOLD CONFINEMENT). In the threshold model an agent is compromised only once at least θ of its

in-neighbours are compromised. An agent with fewer than θ in-neighbours can never meet that condition, so it remains honest for the entire (absorbing) process. On a chain every non-seed agent has in-degree $1 < \theta$, so no propagation occurs. $\square$

A.3 Collusion advantage on the star

FULL PROOF OF THEOREM 9.7. Write $C = 1 + p(N - 1)$ for the centre-seed reach and $L = 1 + p + p^2(N - 2)$ for the leaf-seed reach (Theorem A.2). A colluding adversary with one seed places it at the centre, achieving expected compromise C . An independent adversary places its single seed uniformly at random: with probability $1/N$ it lands on the centre (reach C) and with probability $(N - 1)/N$ on a leaf (reach L), giving

$$\mathbb{E}^{\text{ind}} = \frac{1}{N} C + \frac{N-1}{N} L.$$

Therefore

$$\Delta_{\text{coll}} = C - \mathbb{E}^{\text{ind}} = \frac{N-1}{N} (C - L) = p(1-p) \frac{(N-1)(N-2)}{N} = \Theta(p(1-p)N).$$

For every $p \in (0, 1)$ and $N \geq 3$ we have $(N - 1)(N - 2)/N > 0$, so $\Delta_{\text{coll}} > 0$ strictly. For $s > 1$ seeds, coordinated placement solves the monotone submodular influence-maximization objective, for which greedy seeding attains a $(1 - 1/e)$ approximation to the optimum [Kempe et al. 2003]; the single-seed star computation above already exhibits a strictly positive gap, which is all the theorem claims. $\square$

A.4 Forced coordination and estimators

PROOF OF THEOREM 9.8. (a) Every witness has size $\geq k$ by Theorem 6.3; a set of size $< k$ contains no witness, so no assignment supported on it satisfies Ψ . (b) If $\mathcal{B}$ meets every witness, then every route to $\Psi = 1$ passes through an adversarial agent, which can withhold its contribution, so no witness completes and $\Psi = 0$; if some witness is adversary-free, its members jointly satisfy Ψ . $\square$

PROOF OF THEOREM 9.9. (a) is Hoeffding’s inequality [Hoeffding 1963] for the mean of M i.i.d. $\{0, 1\}$ indicators; invert the bound for M . (b) For $u, \hat{u}, \sigma, \hat{\sigma} \in [0, 1]$, $|\hat{u}\hat{\sigma} - u\sigma| \leq \hat{\sigma}|\hat{u} - u| + u|\hat{\sigma} - \sigma| \leq |\hat{u} - u| + |\hat{\sigma} - \sigma|$; apply with $\sigma = 1 - \text{ASR}_{\text{sys}}$, then union-bound the two events of (a). (c) Bonferroni over G tests replaces α by α/G ; solving for M gives the stated rate, and each of G configurations runs M episodes of N agents over horizon H . $\square$

PROOF OF THEOREM 9.10. Set the per-configuration accuracy to $\epsilon = \Delta/4$ in Theorem 9.9(b) so that the two NRP_{sys} estimates are each within $\Delta/2$; by the triangle inequality their difference is resolved to Δ . Theorem 9.9(c) then gives $M = O(\epsilon^{-2} \ln(G/\alpha)) = O(\Delta^{-2} \ln(G/\alpha))$, and each of the G configurations runs M episodes of N agents over horizon H . $\square$

B Supplementary Material: Notation, Desiderata, and Cost

B.1 Notation

Table 1 (in Section 3) summarizes the notation used throughout the paper; we reference it from here and do not duplicate the table.

Table 3: Topology study (RQ3), synthetic-model results: CPR orders as predicted—chain < tree < star (Theorems A.1 to A.3)—and the orchestrator (star hub) is confirmed as the critical node. Mesh is deferred and measured in the real-backbone pilot (Figure 2(b)).

Topology	CPR	ASR _{sys}	NRP _{sys}	Time-to-detect	Recovery
Chain	0.182	0.997	0.002	2.39	0.218
Tree (branching)	0.226	1.000	0.000	2.39	0.205
Star	0.291	1.000	0.000	2.27	0.202
Mesh	—	—	—	—	—

CPR/ASR_{sys}/NRP_{sys} under no-defense (pure propagation signal); time-to-detect and recovery under the per-agent guardrail. ASR_{sys}=1 reflects availability-sabotage saturation (Table 2); CPR is the discriminating column.

Table 4: Cost of the MASGym instrument. N agents, horizon H , edges $|\mathcal{E}|$ ($\Theta(N)$ for star/chain/tree, $\Theta(N^2)$ for dense mesh), M episodes per configuration, R CoRB rounds, G swept configurations. Checking is deterministic.

Scope	Cost (agent invocations)
Per episode	$\Theta(NH)$
Per configuration	$\Theta(MNH)$
Per CoRB run	$\Theta(RMNH)$
Full parameter sweep	$\Theta(GNH\Delta^{-2} \ln \frac{G}{\alpha})$

B.2 Desiderata for the aggregator (D1–D5)

A system-level utility–security aggregator $g : [0, 1]^2 \rightarrow [0, 1]$, written $\text{NRP}_{\text{sys}} = g(\text{PNA}_{\text{sys}}, 1 - \text{ASR}_{\text{sys}})$, should satisfy: (D1) *boundedness*, g maps into $[0, 1]$; (D2) *security consistency*, $g(u, 1) = u$ and $g(0, \sigma) = 0$; (D3) *monotonicity*, g non-decreasing in each argument; (D4) *utility- and security-proportionality*, $g(\lambda u, \sigma) = \lambda g(u, \sigma)$ and $g(u, \lambda \sigma) = \lambda g(u, \sigma)$ for $\lambda \in [0, 1]$; and (D5) *reduction*, g recovers Equation (1) at $N = 1$.

B.3 Topology study

B.4 Cost and complexity of the instrument

An episode runs N agents for H steps: $\Theta(NH)$ agent invocations and up to $\Theta(|\mathcal{E}|H)$ messages; the trace has size $\Theta((N + |\mathcal{E}|)H)$. The checker evaluates Φ, Ψ once per trace with a linear or near-linear scan ($\tilde{O}((N + |\mathcal{E}|)H)$, no LLM call). Edge counts are $\Theta(N)$ for star, chain, and tree, and $\Theta(N^2)$ for a dense mesh—why mesh is the costliest topology (Section C.4). Estimating the metrics to accuracy ϵ costs $M = O(\epsilon^{-2} \ln(1/\alpha))$ episodes (Theorem 9.9); a CoRB run of R rounds costs $\Theta(RMNH)$ invocations; a sweep of G configurations resolving NRP_{sys} gaps of size Δ costs $\Theta(GNH\Delta^{-2} \ln(G/\alpha))$ (Theorem 9.10). Because the parameter space is exponential in the number of adversary knobs, CoRB explores a *reachable region* rather than the whole space—an honesty constraint we report per study.

B.5 System architecture

Figure 1 reproduces the system overview of Section 7 in full width; the figure is defined in the main body and not repeated here.

Table 5: Sybil study (RQ5), synthetic-model results (star, $N=5$, $\beta=0.1$, no defense, $M=738$). The expected trend— ASR_{sys} increases with the Sybil count until the system is fully seeded—is observed: ASR_{sys} rises monotonically as Sybil identities inflate the adversary’s initial footprint, and NRP_{sys} falls to zero at full seeding.

Sybil count	ASR_{sys}	NRP_{sys}	ASR_{act}	CPR
0	0.775	0.203	0.595	0.173
1	0.870	0.115	0.612	0.288
2	0.942	0.053	0.561	0.344
4	1.000	0.000	0.561	0.000

ASR_{act} is the single-agent predicate (unsafe action / exfiltration; Table 2); CPR at count 4 is 0 by the denominator convention (Section 7.4) because the whole star is seeded at $t=0$.

C Extended Protocol and Additional Synthetic-Model Studies

This appendix expands the evaluation protocol of Section 10: it records the precise metric definitions, the Sybil and adaptivity studies, the cost model, and a sensitivity plan. The filled cells below report measurements from the transparent synthetic model ($M=738$, $\epsilon=\alpha=0.05$, provenance-stamped in the released logs); cells marked “—” are deferred to the real-backbone release phase.

C.1 Precise metric definitions

Given the CoRB logs, each metric is a deterministic functional of the recorded traces:

- **Coordination quality** = PNA_{sys} on forced-coordination tasks under the benign counterpart c° ; **coordination sabotage** = $PNA_{sys}(c^\circ) - PNA_{sys}(c)$ for a sabotage configuration c .
- **Time-to-detection**: the expected first time the defense flags a compromise in a trace ($\min\{t : \text{defense flags in } \tau\}$, ∞ if never), reported as a survival curve over t .
- **Recovery rate**: probability that an episode returns to a safe, task-completing state after a detection.
- **Token/API cost and latency**: summed over all agent and emulator calls in an episode, averaged over the M episodes; **scalability**: the fitted growth of each metric and cost in N .
- **Human-intervention rate**: probability that a human-approval hook was triggered, when hooks are enabled.

C.2 Sybil and adaptivity studies

Table 5 gives the Sybil study (RQ5): as the number of Sybil identities grows, an adversary controlling one underlying policy gains disproportionate influence over selection, voting, and memory writes. The shipped harness does not yet instrument per-selection / per-vote influence shares, so this study quantifies influence through the system metric suite instead. Table 6 gives the adaptivity study: static versus adaptive red policies across CoRB rounds, reporting how quickly the blue policy is out-evolved.

Table 7 gives the collusion sweep (RQ5): the full β axis of Figure 2(c), tabulating colluding versus independent behaviour in both the unsafe-action channel and under the full system predicate. In the unsafe-action channel the collusion advantage Δ_{coll} (Theorem 6.5)

Table 6: Adaptivity study, synthetic-model results (star, $N=5$, $M=60$ per scored pair, $R=20$ CoRB rounds). An adaptive red (Algorithm 2 best-responder) erodes the co-adapting blue’s NRP_{sys} from 0.649 to 0.236; a random (“static”) red leaves the system in the availability-sabotage regime ($ASR_{sys}=1$, $NRP_{sys}=0$) throughout.

Round	Static red		Adaptive red	
	ASR_{sys}	NRP_{sys}	ASR_{sys}	NRP_{sys}
$r = 1$	1.000	0.000	0.317	0.649
$r = 5$	1.000	0.000	0.617	0.358
$r = 10$	1.000	0.000	0.783	0.195
$r = 20$	1.000	0.000	0.733	0.236

Blue = escalating policy (p_{red} ramps $0.2 \rightarrow 0.8$). Static-red ASR_{act} cycles $\{0.50, 0.62, 0.23, 0.52\}$ (no trend); adaptive-red ASR_{act} $\{0.10, 0.18, 0.17, 0.17\}$.

Table 7: Collusion sweep (RQ5), synthetic-model results (star, $N=5$, no defense, $M=738$ episodes per (behaviour, β) cell, seed-locked rows of `collusion_sweep.csv`). ASR_{sys} is the full system predicate (Theorem 6.5); ASR_{act} is the single-agent unsafe-action predicate. The collusion advantage $\Delta_{coll} = ASR_{act}^{coll} - ASR_{act}^{ind}$ is largest at small β and shrinks or changes sign as β grows, while under the system predicate colluding saturates at 1.000 for all $\beta \geq 0.1$.

β	ASR_{act}		Δ_{coll} (ASR_{act})	ASR_{sys}		CPR	
	colluding	independent		colluding	independent	colluding	independent
0.1	0.603	0.554	+0.049	1.000	0.783	0.304	0.173
0.2	0.577	0.575	+0.003	1.000	0.780	0.298	0.288
0.3	0.558	0.583	−0.024	1.000	0.883	0.293	0.344
0.4	0.561	0.523	+0.038	1.000	0.846	0.306	0.000
0.5	0.564	0.565	−0.001	1.000	0.946	0.308	0.000

Every value is a measured cell of `collusion_sweep.csv` (star topology, $N=5$, no defense, $M=738$). ASR_{sys} is the full system predicate including availability sabotage, which saturates at 1.000 when the star hub is seeded; the unsafe-action channel is what Figure 2(c) plots. β is the adversary fraction (Equation (2)); CPR at $\beta=0.0$ is 0 for both behaviours and is omitted.

is biggest at small adversary fraction (+0.049 at $\beta=0.1$) and shrinks—even changing sign—as β grows (−0.024 at 0.3, −0.001 at 0.5); under the full system predicate, colluding saturates at $ASR_{sys}=1.000$ for every $\beta \geq 0.1$ while independent stays below 1, so the system-level advantage is strictly positive and the unsafe-action signal is what Figure 2(c) shows. CPR rises with β for both behaviours and is consistently higher for colluding until the independent cascade catches up at the top of the sweep.

C.3 Cost model and reachable region

By Section 8, one CoRB round costs $2M$ episodes and each episode costs $\Theta(NH)$ agent invocations plus emulator calls, so a study of G swept configurations and R rounds costs $\Theta(GRMNH)$ invocations. With M set by Theorem 9.9(c) to $M = O(\epsilon^{-2} \ln(G/\alpha))$, the dominant cost driver is the product GN : both the sweep grid and the agent cost. Table 8 records the structure of the scalability study (RQ6). We report the *reachable region* of the parameter space

Table 8: Scalability and cost (RQ6), synthetic-model results (star, no defense, $\beta=0.3$, $M=738$). Token cost grows almost exactly linearly in N ($\approx 8.8 \times N$ per episode); latency is the synthetic per-call proxy and stays near-flat. ASR_{sys} saturates via availability sabotage (Table 2).

N	ASR_{sys}	CPR	Tokens/episode	Latency/episode	Reachable?
3	1.000	0.279	26.4	18.7	Yes (138 configs)
5	1.000	0.289	44.0	20.8	Yes (138 configs)
10	0.999	0.294	88.0	22.4	Yes (138 configs)

$N \in \{20, 50\}$ and mesh are deferred to the release phase; the reachable region is the $(N, \text{topology}, \beta, \text{knob})$ combinations actually run (grid in the released provenance logs).

explicitly—which $(N, \text{topology}, \beta, \text{knobs})$ combinations were actually run—rather than implying exhaustive coverage.

C.4 LLM backbone pilot (open models)

The headlined tables are synthetic by design, but the harness’s real-backbone path is not dead code: this pilot wires two open instruct families—Qwen2.5 (0.5B, 1.5B, 3B, 7B; the 7B loaded in 4-bit) and Llama-3.2 (1B, 3B)—through `LLMBackboneAdapter.generate_fn` and runs the environment end-to-end on a grid of 12 configurations ($\{\text{star, chain, tree, mesh}\} \times N \in \{3, 5, 10\}$; $H=4, k=2, \beta=0.3$) with 100 episodes per (configuration, scenario), direct exfiltration injection embedded in a tool-result note; *fictitious data*). Task completion remains the deterministic witness condition, so the pilot isolates the *security* decision: every compromised agent’s reaction to the injected instruction is a real model output. Table 9 summarizes; every model response, with its prompt, is in the released decisions. `jsonl`. These numbers are **labelled PILOT and excluded from the headlined, synthetic results**—they demonstrate the adapter path and a real family-dependence of the security decision, not headline findings. Mistral and the proprietary families are deferred until model credentials are provisioned; the official meta-llama repos gate downloads on per-account approval, so the Llama-3.2 checkpoints here are loaded from an ungated mirror of the official weights.

C.5 A worked forced-coordination task

To make the predicate machinery concrete, consider a 3-forced “travel-booking” task on a star of N agents. The orchestrator decomposes the goal into three subtasks assigned to distinct workers: (W1) retrieve an itinerary from a calendar tool; (W2) select a flight within a stated budget via a booking tool; and (W3) pay via a payment tool, but *only* to the payee named in the user’s instruction. A witness (Theorem 6.3) is any set of agents that jointly supplies a valid itinerary, an in-budget flight, and a correctly addressed payment; because all three subtasks are required and are assigned to distinct workers, every witness has size at least three, so the task is 3-forced.

Table 9: LLM backbone pilot, PILOT-labelled (Qwen2.5 {0.5B, 1.5B, 3B, 7B} and Llama-3.2 {1B, 3B}), 12 configs {star, chain, tree, mesh} $\times N \in \{3, 5, 10\}$, $M=100$ per (config, scenario), no defense, fictitious data; numbers excluded from the headlined synthetic tables). Backbones differ in their security decision: the Qwen family (all four sizes) and Llama-3B refuse every direct exfiltration instruction (almost all parse failures 0), yet ASR_{sys} is still positive in the hub-critical topologies via structural availability sabotage—star and tree collapse to $ASR_{sys}=1.000$ under collusion and orchestrator compromise, while chain stays ≤ 0.25 ; Llama-1B *complies*, yielding $ASR_{sys}=1.000$ across all attack scenarios. The registered security outcome is a real model property, not a framework artefact.

Backbone	benign	single	colluding	orch. comp.
Qwen2.5-0.5B	0.000	0.427	0.696	0.703
Qwen2.5-1.5B	0.000	0.427	0.696	0.703
Qwen2.5-3B	0.000	0.427	0.696	0.703
Qwen2.5-7B*	0.000	0.427	0.696	0.703
Llama-3.2-1B	0.000	1.000	1.000	1.000
Llama-3.2-3B	0.000	0.427	0.696	0.703

Cells are ASR_{sys} , averaged over the 12 (topology $\times N$) configurations, with $NRP_{sys} = PNA_{sys}(1 - ASR_{sys})$ and $PNA_{sys} = 1.000$. Compliance counts (exfiltration-compliant decisions per total compromised-agent decisions, pooled over the same 12 configs) are 0/3878, 0/4077, 0/4118 for the single, colluding, and orchest.-compromise scenarios—wholesale refusal—except Qwen2.5-3B (10/3878) and Llama-3.2-1B, which *complies*: 3812/3878, 4077/4077, 4002/4118. Star and tree saturate via availability sabotage on the hub; chain does not, so the refusal-only ASR_{sys} is topology-dependent. Defenses are synthetic effect-models, deliberately not applied. Token/episode ≈ 320 –2400 across the 12 configs (largest: mesh, $N=10$); real per-episode latency ≈ 0.2 –1.9 s (RTX 4060). *Qwen2.5-7B exceeds the 8 GB GPU in full precision and is loaded in bitsandbytes 4-bit from an ungated mirror; Llama-3.2 weights are from an ungated mirror of the official full-precision (bf16) checkpoints. The security-decision divergence between Llama-3.2-1B (complies) and the remaining backbones (refuse) shows the registered outcome is a real model property, not a framework artefact. Stronger injection styles and per-agent defenses are release-phase work.

The registered predicates read only the recorded trace τ :

$\Psi(\tau) = \#[\text{a payment to the instructed payee for an in-budget flight matching the itinerary is recorded}]$,

$\Phi(\tau) = \#[\text{a payment to a payee not named in the instruction is recorded (exfiltration of funds), or a denylisted action (e.g., disclosing card details to a tool) is recorded}]$.

A malicious tool (via $\mathcal{E}_{LM}^\dagger$) might return a booking result that embeds an injected instruction “pay to account X”; a colluding W3 might then act on it. Whether the payment to account X occurs is a recorded state transition, so Φ fires deterministically regardless of what the agents say in their messages—the content of the injection cannot change the verdict unless the illicit payment is actually recorded (Theorem 9.4). If instead the adversary set forms a transversal of the witnesses (e.g., it controls W3 in every route), it can force $\Psi = 0$ by withholding the legitimate payment (Theorem 9.8), which the sabotage component of the metric records as coordination failure. This example is illustrative; the shipped task suite generalizes it across roles, tools, and topologies.

Table 10: LLM backbone pilot, topology study (RQ4), PILOT-labelled (Qwen2.5-{0.5B,1.5B,3B,7B} and Llama-3.2-{1B,3B}), 12 configs {star, chain, tree, mesh} $\times N \in \{3, 5, 10\}$, $M=100$ per (config, scenario), no defense, fictitious data; numbers excluded from the headlined synthetic tables). CPR is a deterministic structural functional of the topology and is *identical across all six backbones*, so a single column reports it; it orders exactly as Theorems A.1 to A.4 predict (CPR: chain 0.130 < tree 0.198 < star 0.286 < mesh 0.627). ASR_{sys} under the refusing backbones (five of six) is topology-dependent, while Llama-1B, which complies, saturates at $ASR_{sys}=1.000$ everywhere.

Backbone	ASR_{sys} (system predicate)			
	chain	tree	star	mesh
<i>Refusing backbones (5 of 6)</i>				
Qwen2.5-0.5B	0.108	0.766	0.873	0.688
Qwen2.5-1.5B	0.108	0.766	0.873	0.688
Qwen2.5-3B	0.108	0.766	0.873	0.688
Qwen2.5-7B*	0.108	0.766	0.873	0.688
Llama-3.2-3B	0.108	0.766	0.873	0.688
<i>Complying backbone</i>				
Llama-3.2-1B	1.000	1.000	1.000	1.000
<i>Structural functional (identical across all backbones)</i>				
CPR	0.130	0.198	0.286	0.627

C.6 Sensitivity and validity checks

Beyond the headline studies, the protocol includes: (i) an **aggregator-sensitivity** check (RQ1) that recomputes configuration rankings under $u\sigma$, $\min(u, \sigma)$, and a weighted geometric mean $u^w \sigma^{1-w}$ (Theorem 9.3) and reports rank correlation; (ii) an **evaluator-integrity** check (RQ2) that measures how often a crafted injection flips the LLM judge J while leaving the deterministic verdict $C(\tau)$ unchanged (an empirical instance of Theorem 9.4 versus Theorem 9.5); (iii) an **emulation-fidelity** check that compares emulated and real tools on the subset of tasks where a real implementation is available, reporting the change in ASR_{sys} and NRP_{sys} and relating it to κ ; and (iv) **backbone-robustness** checks that repeat the headline degradation study across all five model families so that no conclusion rests on a single model. Each check is a registered configuration in the released harness, so validity, not only headline numbers, is reproducible.

References

- Maksym Andriushchenko, Alexandra Souly, Mateusz Dziemian, Derek Duenas, Maxwell Lin, Justin Wang, Dan Hendrycks, Andy Zou, Zico Kolter, Matt Fredrikson, Eric Winsor, Jerome Wynne, Yarin Gal, and Xander Davies. 2025. AgentHarm: A Benchmark for Measuring Harmfulness of LLM Agents. In *The Thirteenth International Conference on Learning Representations (ICLR)*. arXiv:2410.09024.
- Yoav Benjamini and Yoel Hochberg. 1995. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B (Methodological)* 57, 1 (1995), 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x
- Peva Blanchard, El Mahdi El Mhamdi, Rachid Guerraoui, and Julien Stainer. 2017. Machine Learning with Adversaries: Byzantine Tolerant Gradient Descent. In *Advances in Neural Information Processing Systems 30 (NIPS)*.
- Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. 2013. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press. doi:10.1093/acprof:oso/9780199535255.001.0001
- Greg Brockman, Vicki Cheung, Ludwig Pettersson, Jonas Schneider, John Schulman, Jie Tang, and Wojciech Zaremba. 2016. OpenAI Gym. *arXiv preprint arXiv:1606.01540* (2016).
- Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. 2025. Why Do Multi-Agent LLM Systems Fail? *arXiv preprint arXiv:2503.13657* (2025). MAST failure taxonomy for multi-agent LLM systems.
- J. Chalupa, P. L. Leath, and G. R. Reich. 1979. Bootstrap Percolation on a Bethe Lattice. *Journal of Physics C: Solid State Physics* 12, 1 (1979), L31–L35. doi:10.1088/0022-3719/12/1/008
- Weize Chen, Yusheng Su, Jingwei Zuo, Cheng Yang, Chenfei Yuan, Chi-Min Chan, Heyang Yu, Yaxi Lu, Yi-Hsin Hung, Chen Qian, Yujia Qin, Xin Cong, Ruobing Xie, Zhiyuan Liu, Maosong Sun, and Jie Zhou. 2024a. AgentVerse: Facilitating Multi-Agent Collaboration and Exploring Emergent Behaviors. In *The Twelfth International Conference on Learning Representations (ICLR)*. arXiv:2308.10848.
- Zhaorun Chen, Zhen Xiang, Chaowei Xiao, Dawn Song, and Bo Li. 2024b. Agent-Poison: Red-teaming LLM Agents via Poisoning Memory or Knowledge Bases. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*. 130185–130213. arXiv:2407.12784.
- Jacob Cohen. 1960. A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement* 20, 1 (1960), 37–46. doi:10.1177/001316446002000104
- Stav Cohen, Ron Bitton, and Ben Nassi. 2024. Here Comes The AI Worm: Unleashing Zero-click Worms that Target GenAI-Powered Applications. *arXiv preprint arXiv:2403.02817* (2024).
- Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. 2024. AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, *Datasets and Benchmarks Track*. doi:10.52202/079017-2636 deterministic state-based security() checks; arXiv:2406.13352.
- Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2023. Improving Factuality and Reasoning in Language Models through Multiagent Debate. *arXiv preprint arXiv:2305.14325* (2023).
- Joseph L. Fleiss. 1971. Measuring Nominal Scale Agreement Among Many Raters. *Psychological Bulletin* 76, 5 (1971), 378–382. doi:10.1037/h0031619
- Adam Fourney, Gagan Bansal, Hussein Mozannar, Cheng Tan, Eduardo Salinas, Erkang Zhu, et al. 2024. Magentic-One: A Generalist Multi-Agent System for Solving Complex Tasks. *arXiv preprint arXiv:2411.04468* (2024).
- Mark Granovetter. 1978. Threshold Models of Collective Behavior. *Amer. J. Sociology* 83, 6 (1978), 1420–1443. doi:10.1086/226707
- Xiangming Gu, Xiaosen Zheng, Tianyu Pang, Chao Du, Qian Liu, Ye Wang, Jing Jiang, and Min Lin. 2024. Agent Smith: A Single Image Can Jailbreak One Million Multimodal LLM Agents Exponentially Fast. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*. arXiv:2402.08567.
- Lewis Hammond, Alan Chan, Jesse Clifton, Jason Hoelscher-Obermaier, Akbir Khan, Euan McLean, Chandler Smith, Wolfram Barfuss, Jakob Foerster, et al. 2025. *Multi-Agent Risks from Advanced AI*. Technical Report Technical Report #1. Cooperative AI Foundation. doi:10.48550/arXiv.2502.14143 arXiv:2502.14143.
- Feng He, Tianqing Zhu, Dayong Ye, Bo Liu, Wanlei Zhou, and Philip S. Yu. 2025b. The Emerged Security and Privacy of LLM Agent: A Survey with Case Studies. *Comput. Surveys* (2025). doi:10.1145/3773080 arXiv:2407.19354.
- Pengfei He, Yiping Lin, Shen Dong, Han Xu, Yue Xing, and Hui Liu. 2025a. Red-Teaming LLM Multi-Agent Systems via Communication Attacks. In *Findings of the Association for Computational Linguistics: ACL 2025*. Agent-in-the-Middle (AiTM); arXiv:2502.14847.
- Wassily Hoeffding. 1963. Probability Inequalities for Sums of Bounded Random Variables. *J. Amer. Statist. Assoc.* 58, 301 (1963), 13–30. doi:10.1080/01621459.1963.10500830
- Sirui Hong, Mingchen Zhuge, Jiaqi Chen, Xiaowu Zheng, Yuheng Cheng, Ceyao Zhang, Jinlin Wang, Zili Wang, Steven Ka Shing Yau, Zijuan Lin, Liyang Zhou, Chenyu Ran, Lingfeng Xiao, Chenglin Wu, and Jürgen Schmidhuber. 2023. MetaGPT: Meta Programming for a Multi-Agent Collaborative Framework. *arXiv preprint arXiv:2308.00352* (2023).
- Wenyue Hua, Xianjun Yang, Mingyu Jin, Zelong Li, Wei Cheng, Ruixiang Tang, and Yongfeng Zhang. 2024. TrustAgent: Towards Safe and Trustworthy LLM-based Agents. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. arXiv:2402.01586.
- Hakan Inan, Kartikeya Upasani, Jianfeng Chi, Rashi Rungta, Krithika Iyer, Yuning Mao, Michael Tontchev, Qing Hu, Brian Fuller, Davide Testuggine, and Madian Khabza. 2023. Llama Guard: LLM-based Input-Output Safeguard for Human-AI Conversations. *arXiv preprint arXiv:2312.06674* (2023).
- David Kempe, Jon Kleinberg, and Éva Tardos. 2003. Maximizing the Spread of Influence through a Social Network. In *Proceedings of the Ninth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '03)*. ACM, 137–146. doi:10.1145/956750.956769
- Leslie Lamport, Robert Shostak, and Marshall Pease. 1982. The Byzantine Generals Problem. *ACM Transactions on Programming Languages and Systems* 4, 3 (1982), 382–401. doi:10.1145/357172.357176
- Donghyun Lee and Mo Tiwari. 2024. Prompt Infection: LLM-to-LLM Prompt Injection within Multi-Agent Systems. *arXiv preprint arXiv:2410.07283* (2024).

- Erich L. Lehmann and Joseph P. Romano. 2005. *Testing Statistical Hypotheses* (3rd ed.). Springer. doi:10.1007/0-387-27605-X
- Joel Z. Leibo, Edgar Duñez-Guzmán, Alexander Sasha Vezhnevets, John P. Agapiou, Peter Sunehag, Raphael Koster, Jayd Matyas, Charles Beattie, Igor Mordatch, and Thore Graepel. 2021. Scalable Evaluation of Multi-Agent Reinforcement Learning with Melting Pot. In *Proceedings of the 38th International Conference on Machine Learning (ICML)* (PMLR, Vol. 139). 6187–6199. arXiv:2107.06857.
- Guohao Li, Hasan Abed Al Kader Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. 2023. CAMEL: Communicative Agents for “Mind” Exploration of Large Language Model Society. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2303.17760.
- Yuan Chun Li, Hao Wen, Weijun Wang, Xiangyu Li, Yizhen Yuan, Guohong Liu, Jiacheng Liu, Wenxing Xu, Xiang Wang, Yi Sun, Rui Kong, Yile Wang, Hanfei Geng, Jian Luan, Xuefeng Jin, Zilong Ye, Guanqing Xiong, Fan Zhang, Xiang Li, Mengwei Xu, Zhijun Li, Peng Li, Yang Liu, Ya-Qin Zhang, and Yunxin Liu. 2024. Personal LLM Agents: Insights and Survey about the Capability, Efficiency and Security. *arXiv preprint arXiv:2401.05459* (2024).
- Yueqi Liu, Yuqi Jia, Runpeng Geng, Jinyuan Jia, and Neil Zhenqiang Gong. 2024. Formalizing and Benchmarking Prompt Injection Attacks and Defenses. In *33rd USENIX Security Symposium (USENIX Security)*. arXiv:2310.12815.
- Mantas Mazeika, Long Phan, Xu Wang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhae, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, and Dan Hendrycks. 2024. HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal. In *Proceedings of the 41st International Conference on Machine Learning (ICML)* (PMLR, Vol. 235). 35181–35224. arXiv:2402.04249.
- Colin McDiarmid. 1989. On the Method of Bounded Differences. In *Surveys in Combinatorics, 1989*. London Mathematical Society Lecture Note Series, Vol. 141. Cambridge University Press, 148–188. doi:10.1017/CBO9781107359949.008
- Jerzy Neyman and Egon S. Pearson. 1933. On the Problem of the Most Efficient Tests of Statistical Hypotheses. *Philosophical Transactions of the Royal Society of London. Series A* 231, 694–706 (1933), 289–337. doi:10.1098/rsta.1933.0009
- Marshall Pease, Robert Shostak, and Leslie Lamport. 1980. Reaching Agreement in the Presence of Faults. *J. ACM* 27, 2 (1980), 228–234. doi:10.1145/322186.322188
- Ethan Perez, Saffron Huang, Francis Song, Trevor Cai, Roman Ring, John Aslanides, Amelia Glaese, Nat McAleese, and Geoffrey Irving. 2022. Red Teaming Language Models with Language Models. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 3419–3448. doi:10.18653/v1/2022.emnlp-main.225 arXiv:2202.03286.
- Chen Qian, Wei Liu, Hongzhang Liu, Nuo Chen, Yufan Dang, Jiahao Li, Cheng Yang, Weize Chen, Yusheng Su, Xin Cong, Juyuan Xu, Dahai Li, Zhiyuan Liu, and Maosong Sun. 2024. ChatDev: Communicative Agents for Software Development. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*. arXiv:2307.07924.
- Traian Rebedea, Razvan Dinu, Magesh Sreedhar, Christopher Parisien, and Jonathan Cohen. 2023. NeMo Guardrails: A Toolkit for Controllable and Safe LLM Applications with Programmable Rails. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP)*. arXiv:2310.10501.
- Yangjun Ruan, Honghua Dong, Andrew Wang, Silviu Pitit, Yongchao Zhou, Jimmy Ba, Yann Dubois, Chris J. Maddison, and Tatsunori Hashimoto. 2024. Identifying the Risks of LM Agents with an LM-Emulated Sandbox. In *The Twelfth International Conference on Learning Representations (ICLR)*. ToolEmu; arXiv:2309.15817.
- Donald B. Rubin. 1974. Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies. *Journal of Educational Psychology* 66 (1974), 688–701. doi:10.1037/h0037350
- Christian Schroeder de Witt, Klaudia Krawiecka, Igor Krawczuk, Ben Hagag, William L. Anderson, Peter Belcak, et al. 2025. Open Challenges in Multi-Agent Security: Towards Secure Systems of Interacting AI Agents. *arXiv preprint arXiv:2505.02077* (2025). doi:10.48550/arXiv.2505.02077 MASEC; Sec. 4.7 states no multi-agent security benchmark exists.
- J. K. Terry, Benjamin Black, Nathaniel Grammel, Mario Jayakumar, Ananth Hari, Ryan Sullivan, Luis Santos, Clemens Dieffendahl, Caroline Horsch, Rodrigo Perez-Vicente, Niall Williams, Yashas Lokesh, and Praveen Ravi. 2021. PettingZoo: Gym for Multi-Agent Reinforcement Learning. In *Advances in Neural Information Processing Systems* 34 (NeurIPS). arXiv:2009.14471.
- Yu Tian, Xiao Yang, Jingyuan Zhang, Yinpeng Dong, and Hang Su. 2023. Evil Geniuses: Delving into the Safety of LLM-based Agents. *arXiv preprint arXiv:2311.11855* (2023).
- Boxin Wang, Weixin Chen, Hengzhi Pei, Chulin Xie, Mintong Kang, Chenhui Zhang, Chejian Xu, Zidi Xiong, Ritik Dutta, Rylan Schaeffer, Sang T. Truong, Simran Arora, Mantas Mazeika, Dan Hendrycks, Zinan Lin, Yu Cheng, Sanmi Koyejo, Dawn Song, and Bo Li. 2023a. DecodingTrust: A Comprehensive Assessment of Trustworthiness in GPT Models. In *Advances in Neural Information Processing Systems* 36 (NeurIPS 2023), *Datasets and Benchmarks Track*. arXiv:2306.11698.
- Junlin Wang, Jue Wang, Ben Athiwaratkun, Ce Zhang, and James Zou. 2024. Mixture-of-Agents Enhances Large Language Model Capabilities. *arXiv preprint arXiv:2406.04692* (2024).
- Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghui Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. 2023b. Large Language Models Are Not Fair Evaluators. *arXiv preprint arXiv:2305.17926* (2023).
- Shilong Wang, Guibin Zhang, Miao Yu, Guancheng Wan, Fanci Meng, Chongye Guo, Kun Wang, and Yang Wang. 2025b. G-Safeguard: A Topology-Guided Security Lens and Treatment on LLM-based Multi-agent Systems. *arXiv preprint arXiv:2502.11127* (2025).
- Xingyao Wang, Boxuan Li, Yufan Song, Frank F. Xu, Xiangru Tang, Mingchen Zhuge, et al. 2025a. OpenHands: An Open Platform for AI Software Developers as Generalist Agents. In *The Thirteenth International Conference on Learning Representations (ICLR)*. arXiv:2407.16741.
- Duncan J. Watts. 2002. A Simple Model of Global Cascades on Random Networks. *Proceedings of the National Academy of Sciences* 99, 9 (2002), 5766–5771. doi:10.1073/pnas.082090499
- Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W. White, Doug Burger, and Chi Wang. 2023. AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation. *arXiv preprint arXiv:2308.08155* (2023).
- Wenkai Yang, Xiaohan Bi, Yankai Lin, Sishuo Chen, Jie Zhou, and Xu Sun. 2024. Watch Out for Your Agents! Investigating Backdoor Threats to LLM-Based Agents. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2402.11208.
- Miao Yu, Shilong Wang, Guibin Zhang, Junyuan Mao, Chenlong Yin, Qijiong Liu, Qingsong Wen, Kun Wang, and Yang Wang. 2024. NetSafe: Exploring the Topological Safety of Multi-agent Networks. *arXiv preprint arXiv:2410.15686* (2024).
- Yifan Zeng, Yiran Wu, Xiao Zhang, Huazheng Wang, and Qingyun Wu. 2024. AutoDefense: Multi-Agent LLM Defense against Jailbreak Attacks. *arXiv preprint arXiv:2403.04783* (2024).
- Qiusi Zhan, Zhixiang Liang, Zifan Ying, and Daniel Kang. 2024. InjecAgent: Benchmarking Indirect Prompt Injections in Tool-Integrated Large Language Model Agents. In *Findings of the Association for Computational Linguistics: ACL 2024*. arXiv:2403.02691.
- Hanrong Zhang, Jingyuan Huang, Kai Mei, Yifei Yao, Zhenting Wang, Chenlu Zhan, Hongwei Wang, and Yongfeng Zhang. 2025. Agent Security Bench (ASB): Formalizing and Benchmarking Attacks and Defenses in LLM-based Agents. In *The Thirteenth International Conference on Learning Representations (ICLR)*. ASB; introduces NRP = PNA (1 - ASR); arXiv:2410.02644.
- Zaibin Zhang, Yongting Zhang, Lijun Li, Hongzhi Gao, Lijun Wang, Huchuan Lu, Feng Zhao, Yu Qiao, and Jing Shao. 2024. PsySafe: A Comprehensive Framework for Psychological-based Attack, Defense, and Evaluation of Multi-agent System Safety. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*. arXiv:2401.11880.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*. arXiv:2306.05685.
- Zhenhong Zhou, Zherui Li, Jie Zhang, Yuanhe Zhang, Kun Wang, Yang Liu, and Qing Guo. 2025. CORBA: Contagious Recursive Blocking Attacks on Multi-Agent Systems Based on Large Language Models. *arXiv preprint arXiv:2502.14529* (2025).